EML 6934: Exam 2

March 18, 2026

Factsheet (Personal Reference)

Bayes rules

$$p(x,y) = p(x|y)\,p(y) = p(y|x)\,p(x)$$

Law of total probability

$$p(x) = \int p(x,y)\,dy$$
$$p(x) = \sum_y p(x,y)$$

Law of total probability with conditioning

$$p(x|z) = \sum_y p(y|z)\,p(x|y,z)$$

Corollary of Bayes + law of total probability:

$$p(x) = \int p(x|y)\,p(y)\,dy$$
$$p(x) = \sum_y p(x|y)\,p(y)$$

Independence

$$p(x,y) = p(x)\,p(y) \iff p(x|y) = p(x)$$

Normalization

$$p(x|y) = \underbrace{\frac{1}{\sum_y p(x,y)} p(y|x)}_{\eta}\,p(x)$$
$$= \underbrace{\frac{1}{\sum_y p(y|x)\,p(x)} p(y|x)}_{\eta}\,p(x)$$

Problem 1.1

We are given that

$$\begin{aligned}
P(x_{t+1} = clean | x_t = dirty, u_t = clean) &= 0.6 \\
P(x_{t+1} = dirty | x_t = clean, u_t = clean) &= 0 \\
P(z_t = clean | x_t = dirty) &= 0.2 \\
P(z_t = clean | x_t = clean) &= 0.6 \\
z_{t+1} &= clean
\end{aligned}$$

The robot:

1. Lands on a new tile
2. Cleans it (without sensing) at t

3. Then senses if it is clean or not at $t+1$

We assume no prior knowledge of the floor tile we're over:

$$P\left(x_t = clean\right) = c \in [0,1]$$

We want a plot of:

$$P\left(x_{t+1} = clean | z_{t+1} = clean\right).$$

Use the Bayes filter (Table 2.1 from Sebastian Thrun's book). Step 0: initialize the belief

$$bel\left(x_t = clean\right) = c$$

Step 1: compute the time update (prediction a.k.a posterior belief before the measurement).

$$\begin{aligned}
\overline{bel}\left(x_{t+1} = clean\right) &= \int p\left(x_{t+1} = clean | u_{t+1} = clean, x_t = unknown\right) bel\left(x_t\right) dx_t \\
&= p\left(x_{t+1} = clean | u_{t+1} = clean, x_t = clean\right) p\left(x_t = clean\right) \\
&+ p\left(x_{t+1} = clean | u_{t+1} = clean, x_t = dirty\right) p\left(x_t = dirty\right) \\
&= p\left(x_{t+1} = clean | u_{t+1} = clean, x_t = clean\right) p\left(x_t = clean\right) \\
&+ p\left(x_{t+1} = clean | u_{t+1} = clean, x_t = dirty\right) p\left(x_t = dirty\right) \\
&= 1 \cdot c + 0.6 \cdot (1-c) \\
&= 0.4c + 0.6
\end{aligned}$$

Step 2: compute the measurement update (posterior belief after the measurement). Use the law of total probability and Bayes rule.

$$\begin{aligned}
bel\left(x_{t+1} = clean\right) &= \underbrace{\frac{1}{p\left(z_{t+1} = clean\right)}}_{\eta} p\left(z_{t+1} = clean | x_{t+1} = clean\right) \overline{bel}\left(x_{t+1} = clean\right) \\
&= \frac{1}{\sum_x p\left(z_{t+1} = clean, x_{t+1}\right)} p\left(z_{t+1} = clean | x_{t+1} = clean\right) \overline{bel}\left(x_{t+1} = clean\right) \\
&= \frac{1}{\sum_x p\left(z_{t+1} = clean | x_{t+1}\right) p\left(x_{t+1}\right)} p\left(z_{t+1} = clean | x_{t+1} = clean\right) \overline{bel}\left(x_{t+1} = clean\right) \\
&= \frac{p\left(z_{t+1} = clean | x_{t+1} = clean\right) \overline{bel}\left(x_{t+1} = clean\right)}{p\left(z_{t+1} = clean | x_{t+1} = clean\right) p\left(x_{t+1} = clean\right) + p\left(z_{t+1} = clean | x_{t+1} = dirty\right) p\left(x_{t+1} = dirty\right)} \\
&= \frac{p\left(z_{t+1} = clean | x_{t+1} = clean\right) \overline{bel}\left(x_{t+1} = clean\right)}{p\left(z_{t+1} = clean | x_{t+1} = clean\right) p\left(x_{t+1} = clean\right) + p\left(z_{t+1} = clean | x_{t+1} = dirty\right) \left(1 - p\left(x_{t+1} = clean\right)\right)} \\
&= \frac{p\left(z_{t+1} = clean | x_{t+1} = clean\right) \overline{bel}\left(x_{t+1} = clean\right)}{p\left(z_{t+1} = clean | x_{t+1} = clean\right) \overline{bel}\left(x_{t+1} = clean\right) + p\left(z_{t+1} = clean | x_{t+1} = dirty\right) \left(1 - \overline{bel}\left(x_{t+1} = clean\right)\right)} \\
&= \frac{0.6 \cdot (0.4c + 0.6)}{0.6 \cdot (0.4c + 0.6) + 0.2\left(1 - (0.4c + 0.6)\right)} \\
&= \frac{0.24c + 0.36}{0.24c + 0.36 + 0.2 - 0.2\left(0.4c + 0.6\right)} \\
&= \frac{0.24c + 0.36}{0.24c + 0.36 + 0.2 - 0.08c - 0.12} \\
&= \frac{0.24c + 0.36}{0.16c + 0.44}
\end{aligned}$$

Figure 1 shows the above function.

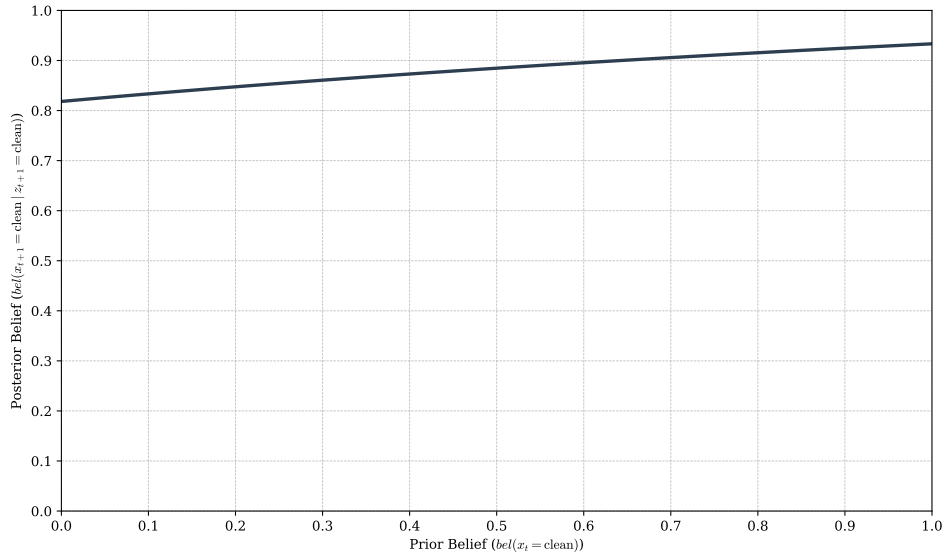

Figure 1: Posterior belief after applying Bayer's filter to the robot vacuum in the described case.

Problem 1.2

Consider the time-invariant, discrete-time, nonlinear state space system (from Thrun, Problem 5.40)

$$\dot{s} = f\left(s_k, u_k\right),$$
$$f\left(s, u\right) \triangleq \begin{bmatrix} s_1 + u_2 \cos\left(s_3 + u_1\right) \\ s_2 + u_2 \sin\left(s_3 + u_1\right) \\ s_3 + u_1 + u_3 \end{bmatrix}$$

where $u \triangleq \begin{bmatrix} u_1^\top & u_2^\top & u_3^\top \end{bmatrix}^\top = \begin{bmatrix} \delta_{\text{rot1}}^\top & \delta_{\text{trans}}^\top & \delta_{\text{rot2}}^\top \end{bmatrix}^\top \in \mathbb{R}^3$ and $s \triangleq \begin{bmatrix} x^\top & y^\top & \theta^\top \end{bmatrix}^\top \in \mathbb{R}^3$. Let the corresponding measurement model be

$$h\left(s\right) \triangleq \arctan\left(\frac{\ell_y - s_2}{\ell_x - s_1}\right) - s_3$$

Take the Jacobians.

$$\begin{aligned} \frac{df}{du} &= \begin{bmatrix} \frac{df_1}{du_1} & \frac{df_1}{du_2} & \frac{df_1}{du_3} \\ \frac{df_2}{du_1} & \frac{df_2}{du_2} & \frac{df_2}{du_3} \\ \frac{df_3}{du_1} & \frac{df_3}{du_2} & \frac{df_3}{du_3} \end{bmatrix} \\ &= \begin{bmatrix} -u_2 \sin\left(s_3 + u_1\right) & \cos\left(s_3 + u_1\right) & 0 \\ u_2 \cos\left(s_3 + u_1\right) & \sin\left(s_3 + u_1\right) & 0 \\ 1 & 0 & 1 \end{bmatrix} \end{aligned}$$

and

$$\begin{aligned} \frac{df}{ds} &= \begin{bmatrix} \frac{df_1}{ds_1} & \frac{df_1}{ds_2} & \frac{df_1}{ds_3} \\ \frac{df_2}{ds_1} & \frac{df_2}{ds_2} & \frac{df_2}{ds_3} \\ \frac{df_3}{ds_1} & \frac{df_3}{ds_2} & \frac{df_3}{ds_3} \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & -u_2 \sin\left(s_3 + u_1\right) \\ 0 & 1 & u_2 \cos\left(s_3 + u_1\right) \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

Algorithm Extended_Kalman_filter($\mu_{t-1}, \Sigma_{t-1}, u_t, z_t$):

$\bar{\mu}_t = g(u_t, \mu_{t-1})$

$\bar{\Sigma}_t = G_t \, \Sigma_{t-1} \, G_t^T + R_t$

$K_t = \bar{\Sigma}_t \, H_t^T (H_t \, \bar{\Sigma}_t \, H_t^T + Q_t)^{-1}$

$\mu_t = \bar{\mu}_t + K_t(z_t - h(\bar{\mu}_t))$

$\Sigma_t = (I - K_t \, H_t) \, \bar{\Sigma}_t$

return μ_t, Σ_t

Figure 2: EKF Algorithm

and

$$\begin{aligned}
\frac{dh}{ds} &= \begin{bmatrix} \frac{dh_1}{ds_1} \\ \frac{dh_1}{ds_2} \\ \frac{dh_1}{ds_3} \end{bmatrix} \\
&= \begin{bmatrix} \frac{1}{1+\left(\frac{\ell_y - s_2}{\ell_x - s_1}\right)^2} \cdot (\ell_y - s_2)\,(\ell_x - s_1)^{-2} \\ \frac{1}{1+\left(\frac{\ell_y - s_2}{\ell_x - s_1}\right)^2} \cdot -(\ell_x - s_1)^{-1} \\ -1 \end{bmatrix} \\
&= \begin{bmatrix} \frac{\ell_y - s_2}{(\ell_x - s_1)^2 + (\ell_y - s_2)^2} \\ \frac{s_1 - \ell_x}{(\ell_x - s_1)^2 + (\ell_y - s_2)^2} \\ -1 \end{bmatrix}
\end{aligned}$$

Thus, at a given operating point $(s, u) \in \mathbb{R}^3 \times \mathbb{R}^3$, we have the linear system

$$\begin{aligned}
s_k &= \left(\frac{df}{ds}\right)_k \Delta s_k + \left(\frac{df}{du}\right)_k \Delta u_k + \varepsilon \\
y_k &= \left(\frac{dh}{ds}\right)_k s_k + \delta
\end{aligned}$$

where ε and δ are zero-mean Gaussian disturbances and noise, respectively, each with known covariances.

Problem 2.1a

The pseudo-code for the EKF algorithm is given in Figure 2 (Sebastian Thrun's Probabilistic Robots book, Table 2.1) where the variables above are defined as follows:

- $x_t \in \mathbb{R}^n$ is the state at time t
- $u_t \in \mathbb{R}^m$ is the control input at time t
- $z_t \in \mathbb{R}^p$ is the measurement at time t
- $h : \mathbb{R}^n \to \mathbb{R}^p$ is the known measurement model
- $f : \mathbb{R}^n \to \mathbb{R}^n$ is the known dynamical model
- $R_t \in \mathbb{R}^{n \times n}$ is a symmetric positive-definite matrix representing the known covariance of the model uncertainty
- $Q_t \in \mathbb{R}^{p \times p}$ is a symmetric positive-definite matrix representing the known covariance of the measurement process
- $\mu_{t-1} \in \mathbb{R}^n$ and $\Sigma_{t-1} \in \mathbb{R}^{n \times n}$ are the mean and covariance (which are sufficient to represent a Gaussian prior belief)
- $\overline{\mu}_t \in \mathbb{R}^n$ and $\overline{\Sigma}_t \in \mathbb{R}^{n \times n}$ are the mean and covariance after the prediction step (but before the measurement/correction update)
- $H_t \triangleq \frac{dh}{dx}\Big|_{\mu_{t-1}, u_t} \in \mathbb{R}^{p \times n}$ is the Jacobian matrix for the measurement model h with respect to the state x at time t
- $G_t \triangleq \frac{\partial f}{\partial x}\Big|_{\mu_{t-1}, u_t} \in \mathbb{R}^{n \times n}$ is the Jacobian matrix of the dynamical model f with respect to the state x at time t

```
Algorithm Particle_filter(X_{t-1}, u_t, z_t):
    X̄_t = X_t = ∅
    for m = 1 to M do
        sample x_t^[m] ~ p(x_t | u_t, x_{t-1}^[m])
        w_t^[m] = p(z_t | x_t^[m])
        X̄_t = X̄_t + ⟨x_t^[m], w_t^[m]⟩
    endfor
    for m = 1 to M do
        draw i with probability ∝ w_t^[i]
        add x_t^[i] to X_t
    endfor
    return X_t
```

Figure 3: PF Algorithm

- $\mu_t \in \mathbb{R}^n$ and $\Sigma_t \in \mathbb{R}^{n \times n}$ are the mean and covariance after the measurement/correction update
- $K_t \in \mathbb{R}^{n \times p}$ is optimal Kalman gain matrix at time t
- $I \in \mathbb{R}^{n \times n}$ is the $\mathbb{R}^{n \times n}$ identity matrix (although this line should really be implemented with Jospeh form for better long-term numerical stability)

Problem 2.2a

The pseudo-code for the particle filter algorithm is given in Figure 3 (Sebastian Thrun's Probabilistic Robots book, Table 4.3) where the variables above are defined as follows:

- $x_t \in \mathbb{R}^n$ is the state at time t
- $u_t \in \mathbb{R}^m$ is the control input at time t
- $z_t \in \mathbb{R}^p$ is the measurement at time t
- $M \in \mathbb{Z}_{\geq 1}$ is the total number of particles used in the filter
- $\mathcal{X}_{t-1}$ is the prior set of $M \in \mathbb{Z}_{\geq 1}$ particles representing the belief distribution at time $t-1$
- m and i are integer indices used to iterate through the particles
- $x_t^{[m]} \in \mathbb{R}^n$ is the specific state hypothesis of the mth particle at time t
- $w_t^{[m]} \in \mathbb{R}_{\geq 0}$ is the importance weight of the mth particle at time t
- $\overline{\mathcal{X}}_t$ is the intermediate proposed set of particles and their corresponding weights after the prediction step (but before the resampling/correction update)
- $\mathcal{X}_t$ is the posterior set of M particles representing the updated belief distribution at time t (after the resampling step)

Problem 2.1b

The results for the EKF are given in Figure 4.

Problem 2.2b

The results for the PF are given in Figure 5.

Source Code

I do not own MATLAB (and since UF provides no license), I cannot use the starter code, but the codebase used to generate these plots is public and available at: https://github.com/mgardenswartz/eml6934-exam-2.

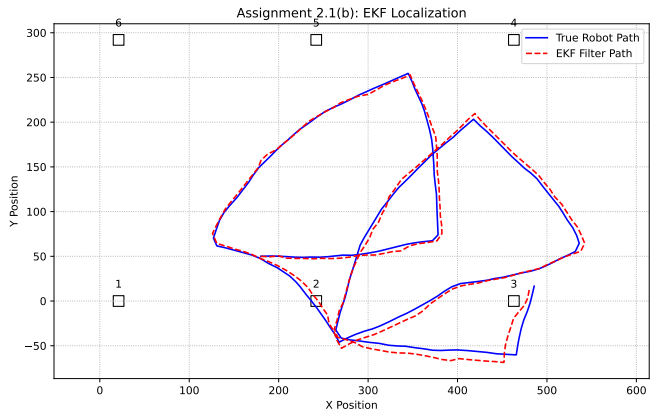

Figure 4: EKF Algorithm

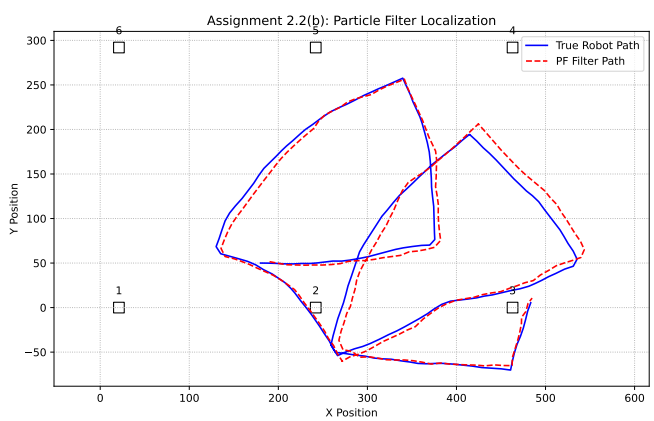

Figure 5: PF Algorithm